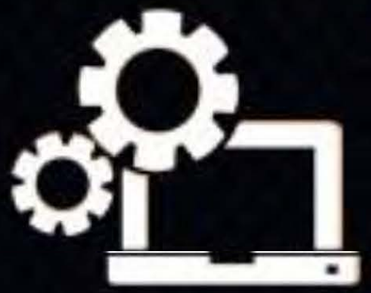
CS & IT ENGINEERING

DIGITAL LOGIC

Chapter : Logic Gates



Lecture No. 2



By - CHANDAN SIR

Recap of Previous Lecture



Topic NOT gate

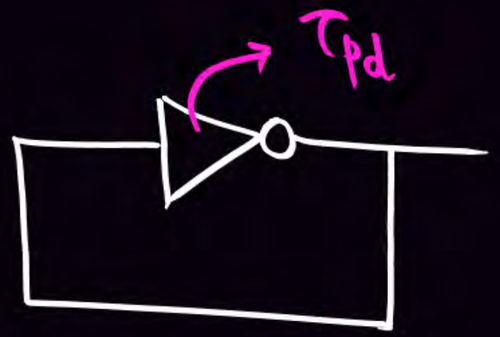




Topics to be Covered

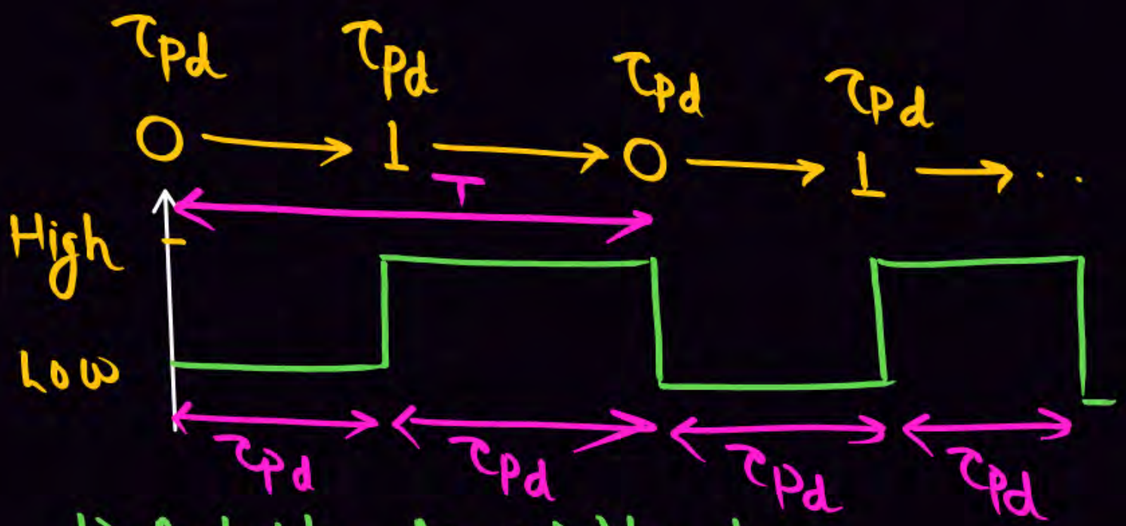


- AND GATE
- OR GATE
- Question Practice
- Discussion



$$T = 2\tau_{pd}$$

$$f = \frac{1}{T} = \frac{1}{2\tau_{pd}}$$



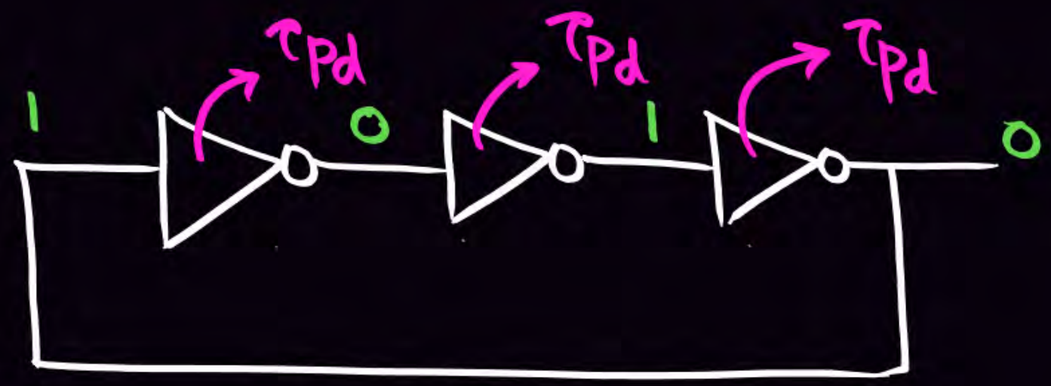
1) Astable Multivibrator

2) Square wave generator

3) Clock generator

4) Free Runing circuit

5) Ring oscillator



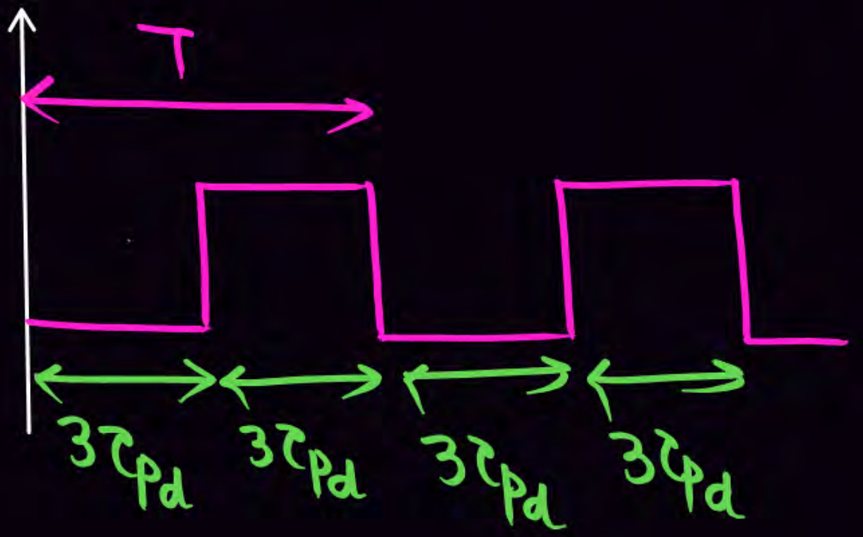
$3\tau_{pd}$ $3\tau_{pd}$ $3\tau_{pd}$

0 → 1 → 0 → 1 →

$T = 6\tau_{pd}$

$f = \frac{1}{6\tau_{pd}}$

$f = \frac{1}{2 \times 3 \times \tau_{pd}}$

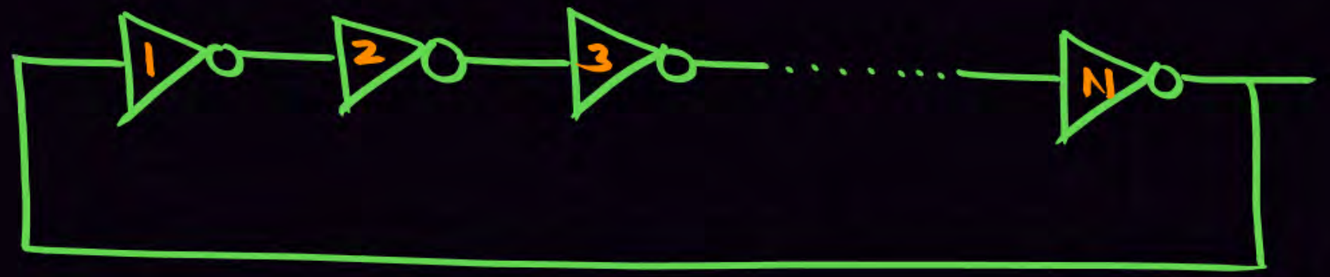


$$T = 2 \times N \times \tau_{pd}$$

$$f = \frac{1}{2 \times N \times \tau_{pd}}$$

N → Number of NOT GATE in Loop.





$N \rightarrow$ EVEN	$N \rightarrow$ Odd
$f = 0\,Hz$	$f = \frac{1}{2N \times \tau_{pd}}$

Q If all the NOT GATES are identical & having $\tau_{Pd} = 2\,\mu s$ Then find the freq. of generated waveform y.



(A) 0 Hz

(B) 500 KHz

~~(C)~~ 50 KHz

(D) None

$N = 5$

$$f = \frac{1}{2N \times \tau_{Pd}}$$

$$f = \frac{1}{2 \times 5 \times 2 \times 10^{-6}} \left(\frac{1}{\text{sec}}\right) \rightarrow \text{Hz}$$

$$f = \frac{10^6}{20}\,\text{Hz} = \frac{50 \times 10^3}{2\cancel{0}}\,\text{Hz}$$

$$= 50\,\text{KHz}$$

Q If all the NOT GATES are identical & having $\tau_{Pd} = 5ps$ Then find the freq. of generated waveform y.

$\tau_{Pd} = 5ps \rightarrow 10^{-12}$

$10^6 Hz \rightarrow MHz$

$10^9 Hz \Rightarrow GHz$



$f = 20\ GHz$

(A) $f = 10\ KHz$

(B) $f = 20\ KHz$

(C) $f = 10\ MHz$

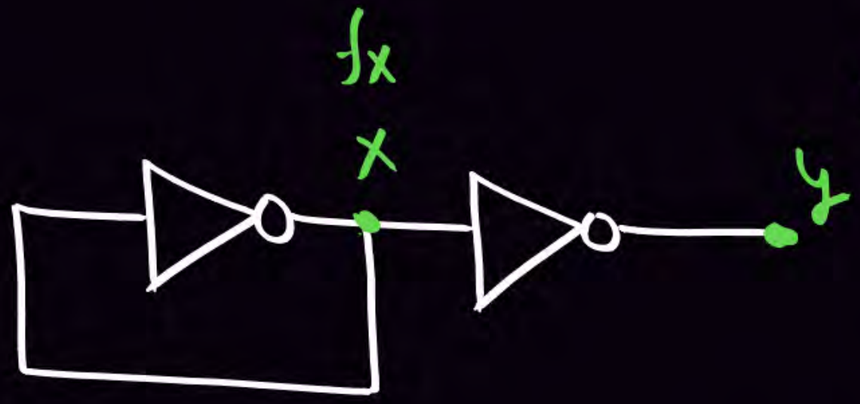
(D) None

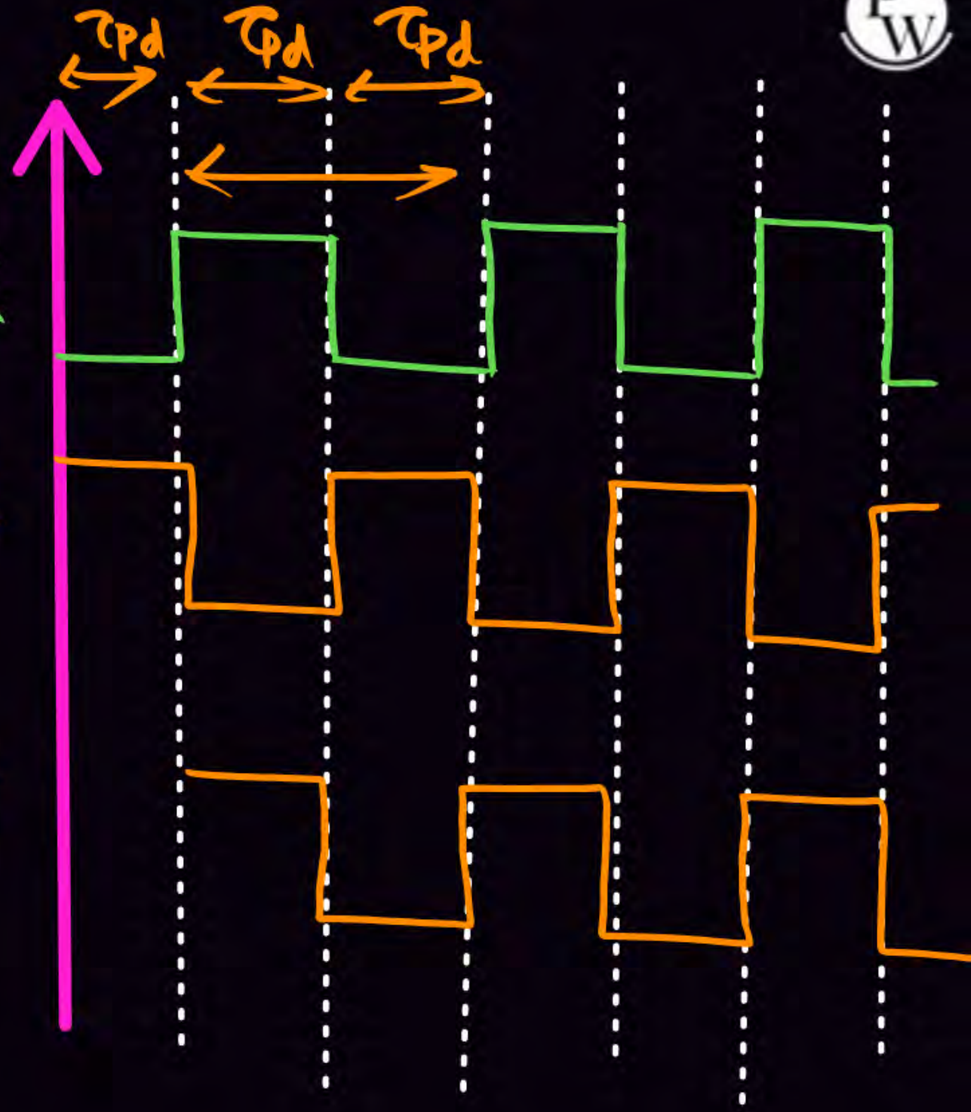
$$f = \frac{1}{2 \times N \times \tau_{Pd}}$$

$$f = \frac{1}{2 \times 5 \times 5 \times 10^{-12}}\ Hz$$

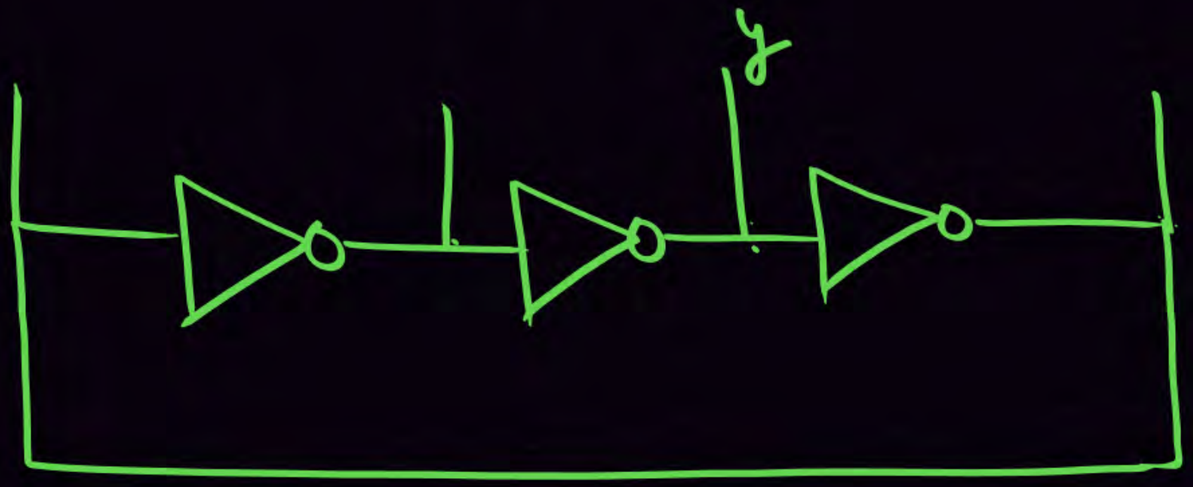
$$f = \frac{10^{12}}{50}\ Hz = \frac{\cancel{100}^{20}}{\cancel{5}0} \times 10^9\ Hz$$

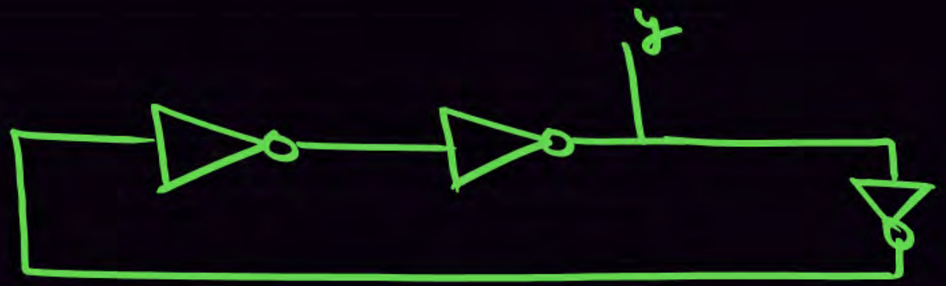
2

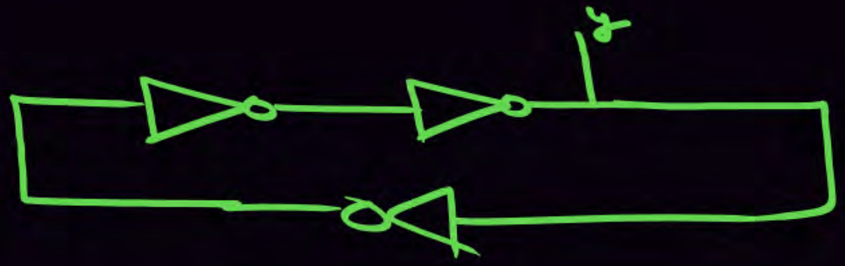


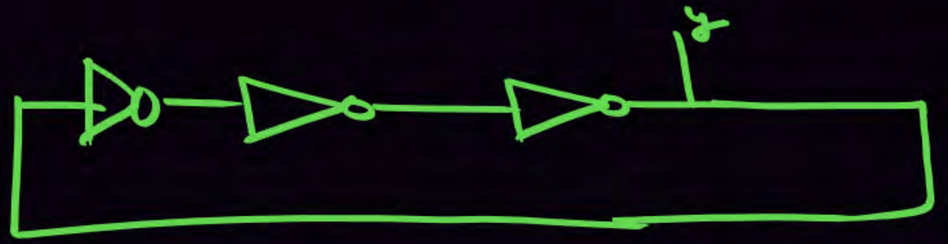


Astable Multivibrator


y


y


y


y



Topic : AND & OR GATE

AND (Intersection)



$0 \cdot 0 = 0$

$0 \cdot 1 = 0$

$1 \cdot 0 = 0$

$1 \cdot 1 = 1$

$1 \cdot A = A$

$A \cdot A = A$

$A \cdot 0 = 0$

$A \cdot \bar{A} = 0$

$1 \cdot A = A$

$1 + A = 1$

OR (Union)



$0 + 0 = 0$

$0 + 1 = 1$

$1 + 0 = 1$

$1 + 1 = 1$

$A + A = A$

$A + 0 = A$

$A + \bar{A} = 1$

$1 + A = 1$

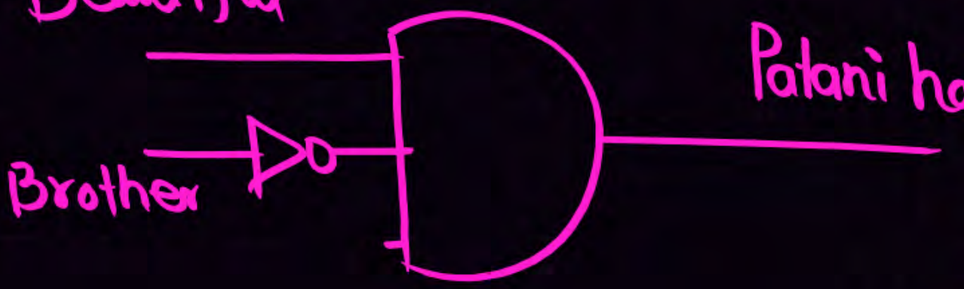
$1 + A = 1$

$1 + AB = 1$

$1 + \text{anything} = 1$

$A \cdot A = A$

2
+2



1
2
3
4
5
:



Topic : AND GATE

1. Symbol

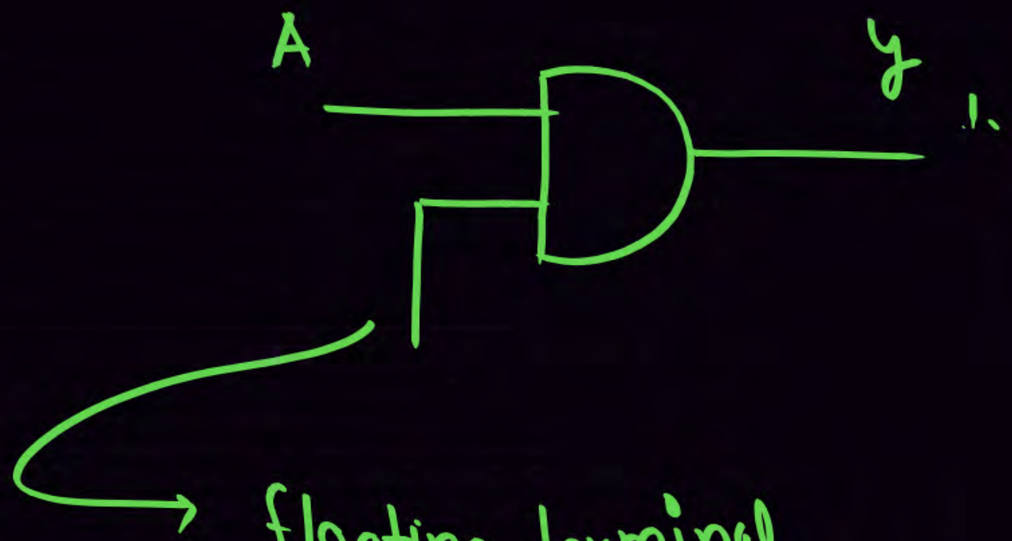


2. Symbol

2. Truth table

→ Relation b/w inputs & outputs

Truth Table		
A	B	Y = A·B
0	0	0
0	1	0
1	0	0
1	1	1



floating terminal

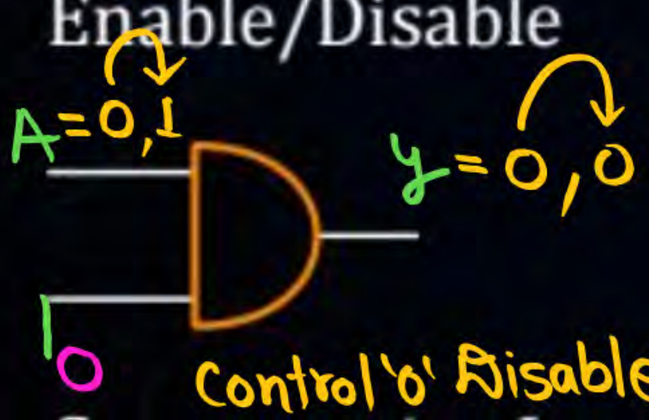
control input

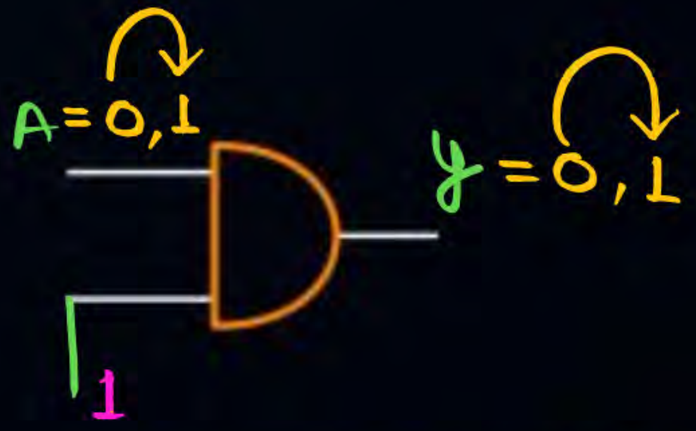
Strobe



Topic : AND GATE

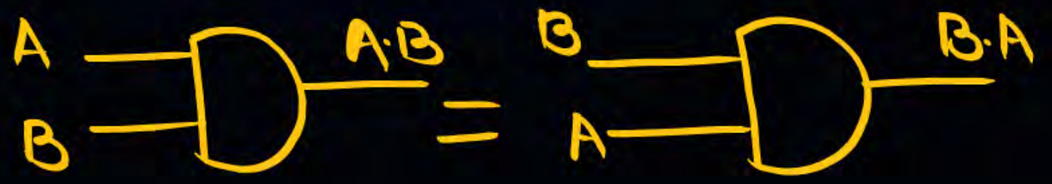
3. Enable/Disable





4. Commutative Law

Control '1' Enable



$A.B = B.A$

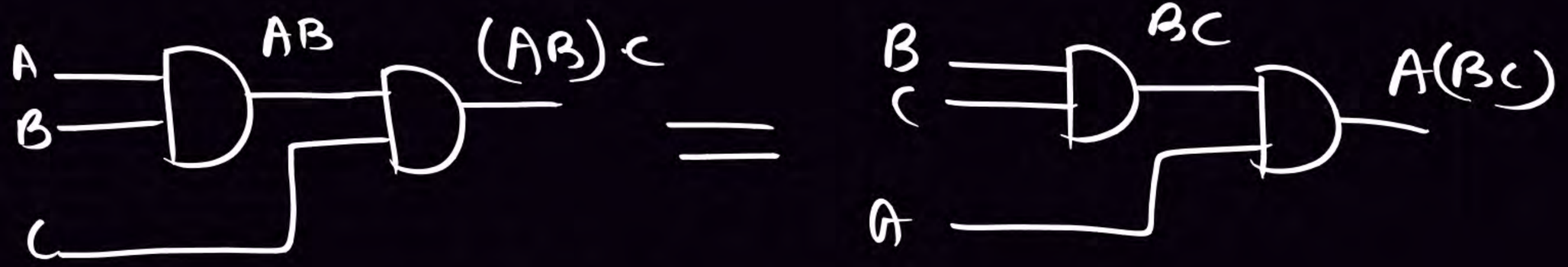
Associative Law

$A.B.C = (A.B).C$

$$(A \cdot B) \cdot C \quad = \quad A \cdot (B \cdot C)$$

Associative Law ✓

$A \cdot B = B \cdot A$ Commutative Law ✓



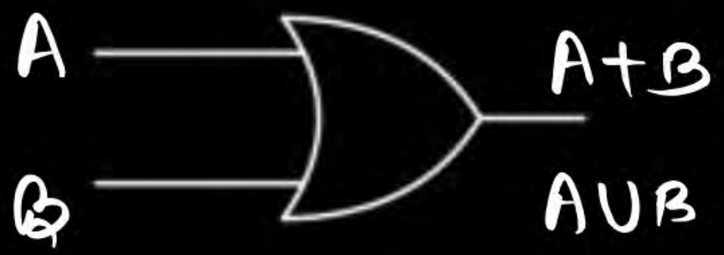
$(AB) \cdot C = A \cdot (BC)$

Associative Law ✓



Topic : OR GATE

1. Symbol

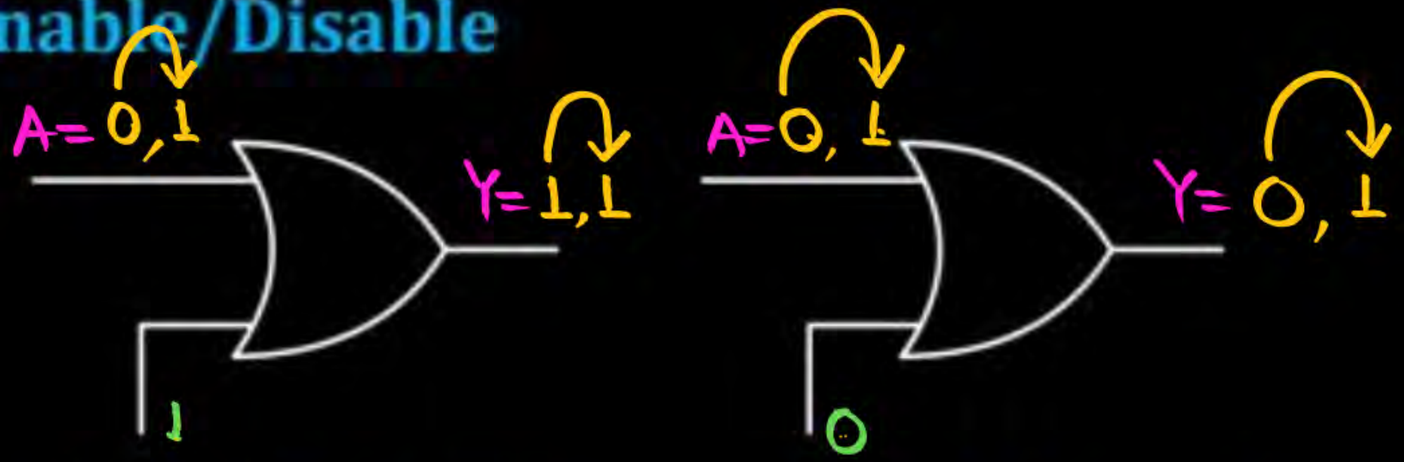


2. Truth Table

A	B	Y = A + B
0	0	0
0	1	1
1	0	1
1	1	1

Topic : OR GATE

3. Enable/Disable



Control '1' Disable

Control '0' Enable

Commutative Law ✓

$$A + B = B + A$$



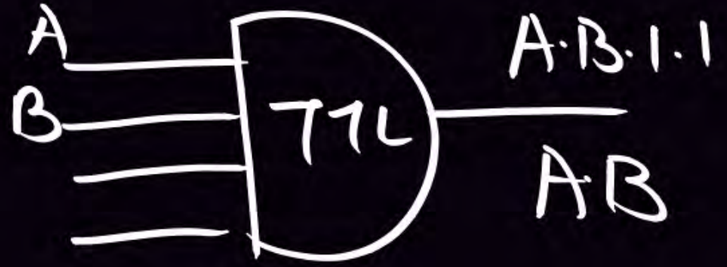
Associative Law ✓

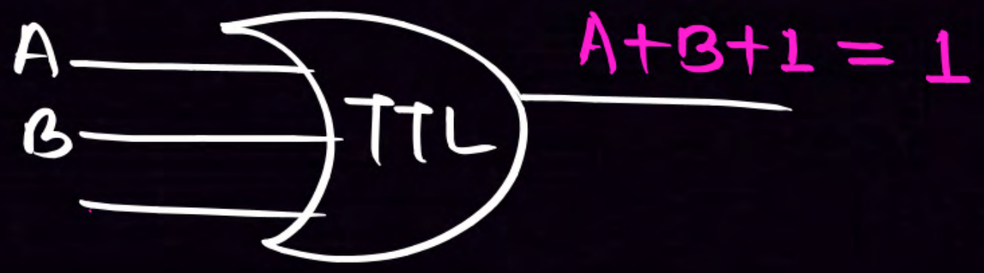
$$(A + B) + C = A + (B + C)$$

TTL → Transistor Transistor Logic

↳ When Logic are designed by TTL Then floating terminal always acts as high.



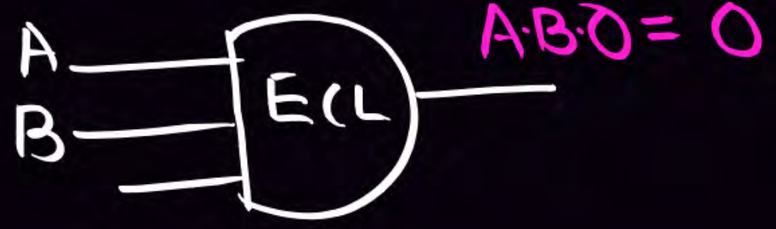



A
B
TTL
A+B+1 = 1

ECL → Emitter coupled Logic

→ When Logic are designed by ECL floating terminal acts as Low.

Ex.



Ex



Topic : OR GATE

Q If logic are designed TTL then o/p y = ?



(A) 0

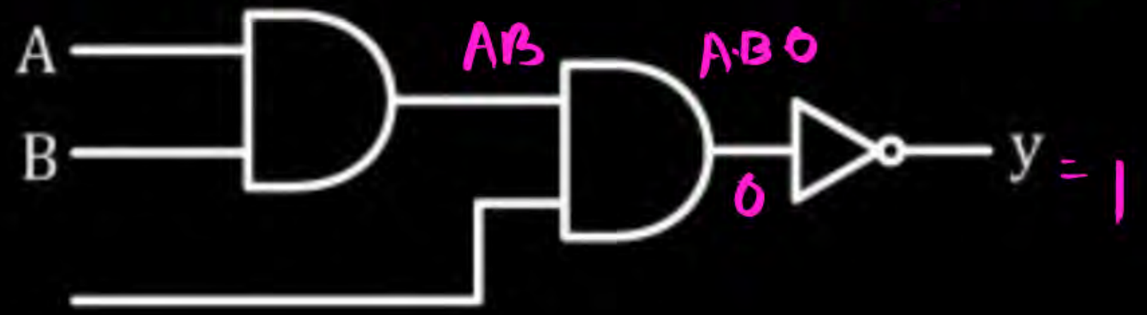
(B) AB

(C) $\overline{AB}$

(D) Mujhe nahi ata hai

Topic : OR GATE

Q If logic are designed ECL then o/p y = ?



(A) 0

(B) AB

(C) $\overline{AB}$

(D) 1

#Q. Consider the given logic circuit with the inputs A, B and C, then f(A, B, C) will be-

A ✗



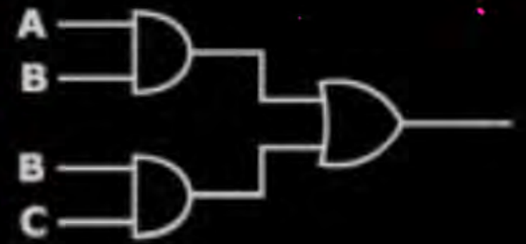
B

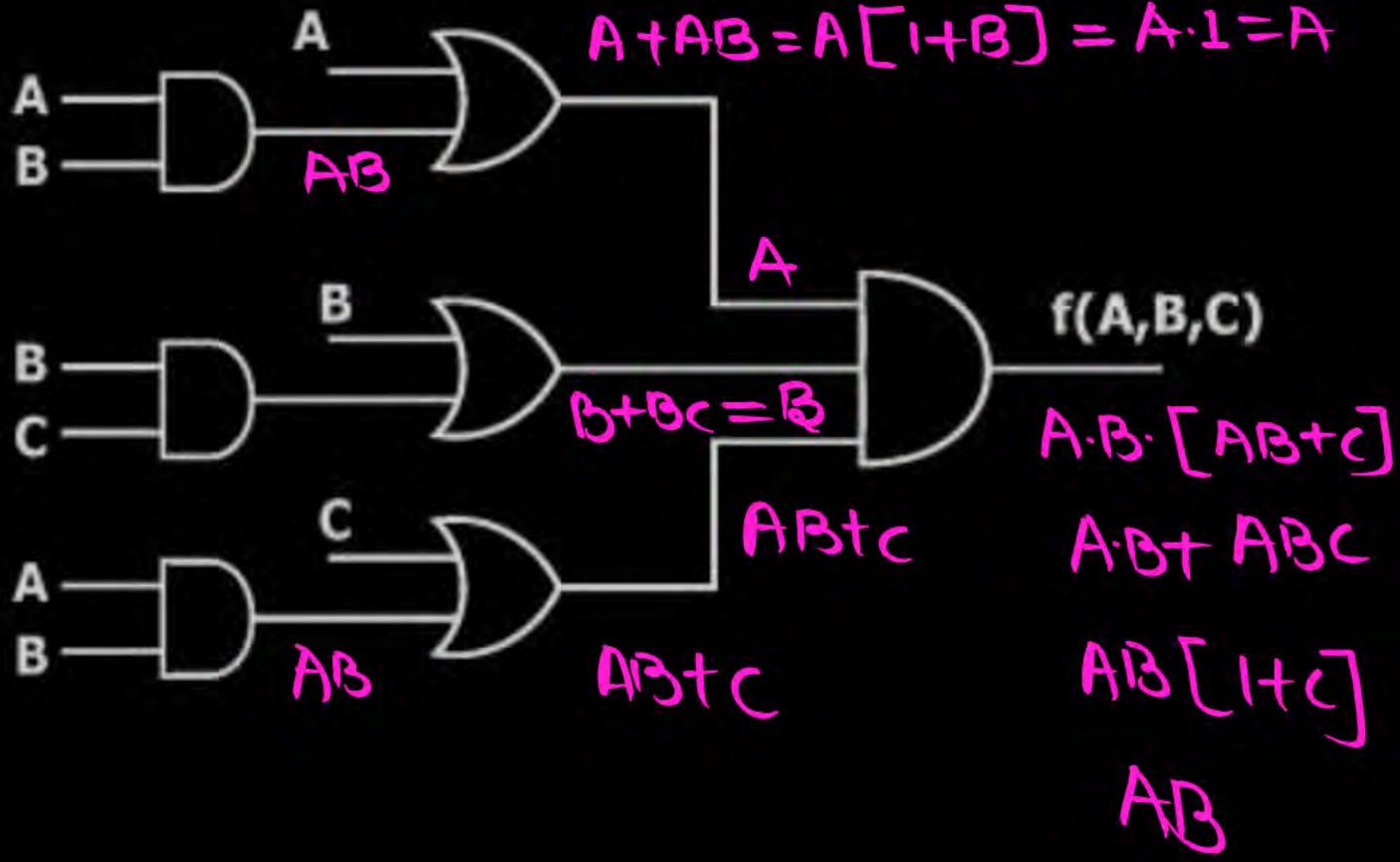


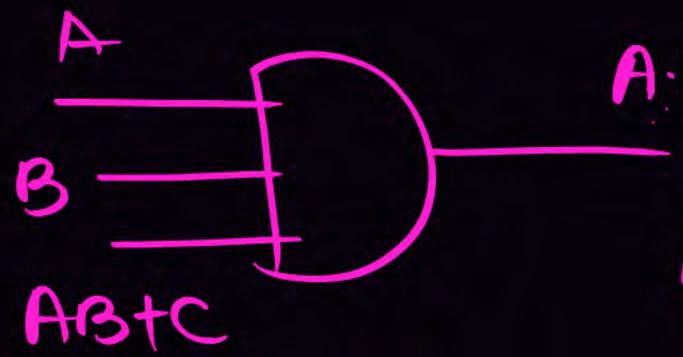
C ✗



D ✗







$A \cdot B \cdot \{AB + C\}$

$AB \cdot AB + AB \cdot C$

$AB + ABC$

$AB[1 + C] = AB \cdot 1$

$= AB$

$=$



2 Min Summary

- AND GATE
- OR GATE
- Question Practice
- Discussion

Thank You GW Soldiers!

